

Automatic Analysis of News Video Interviews

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Abstract—With the increasing volume of video content in the media, people are constantly being exposed to video content. Due to this vast content, it may be likely that we might not have enough time to fully analyse the video content we have watched. Therefore, it would be ideal for a user to have a tool to quickly analyse the full news video so that this could give a better overview of the video.

In this report, a system will be created to carry out automatic news video analysis. This system would make use of various computer vision techniques so that key information would be extracted from the videos. The data then can be viewed by the user to better understand that news video.

The created system managed to identify people by either reading their names from the news video or matching them from a database. This process was fully unsupervised, and if needed will also update the database with new individuals identified.

Index Terms—computer vision, news analysis

I. INTRODUCTION

In a world where information is relatively easy to access, we may become overloaded with information [1] such that we are not able to process all of it which may lead to non-optimal decisions in the future. Specifically, when it comes to media, it is important that we stay up-to-date [2] while also retaining a non-biased view of the content we consume [3]. Since video content in media has been rising, this might be a concern which should be analysed and evaluated accordingly.

To address this problem, tools can be created to give users an insight into what the video contains. Such tools help users in identifying some key information, which may include a summary of the news, and also attempts to make users more aware of the content. Although a number of tools have been created to address specific problems [4], [5], [6], [7], not many addresses and analyses the people shown in the news video.

When it comes to news videos these are not always structured, and if they are these might not have the same format as other news sources [6]. Due to this, it may not always be easy to create such tools to automatically analyse news videos, since some videos require more specific approaches than others. Another challenge that comes with analysing people in the videos is the number of different random people that can appear on the news which can also reappear at different points as well.

A. Proposed Solution

In this study, a system is created to analyse the people that appear in news videos. This system is able to identify any person and detect the time stamps of occurrence as well as the duration. Since usually in news videos the name of the speaking individual is shown, this can be used to extract the individual's name and match the face within a database, or

add to an existing one. An important feature of the system is that it should be unsupervised so that there would not be any need for any manual or supervised interactions, and thus could keep learning on its own. Such a system would make use of different computer vision techniques, namely, face recognition and optical character recognition (OCR) to read the name of the individual.

B. Aims and Objectives

The aim of this study is to develop a method to analyse the news videos which are released on a daily bases. This analysis is done so to assist the user with any insight that could be overlooked. Finally, the user should be presented with an insight into the individuals that appeared in the news video. This aim will be achieved by following these objectives:

- Extract new videos
- Create a system to recognise faces, matches them with a database, and add them to the database if new.
- Extract the name of individuals from the news video
- Mark people's timestamp occurrences.
- Present the user with a report of the people who were shown in the news video, the duration, and also start and end timestamps of occurrences.

II. BACKGROUND

This chapter will provide a short background regarding the techniques used within the created system for news video analysis. To identify and match unique individuals, within the video analysis, Face Recognition will be used. Problems with Face Recognition arise, especially in video format and these will be discussed as well as some solutions to such problems. Moreover, Optical Character Recognition will be used to read the individual names, and the techniques used will be covered. Finally, to identify bounds between different individuals and other non-interview clips, Scene Detection was used. Different algorithms for scene detection will also be covered since there exist numerous different algorithms which are better in some cases than others.

A. Face Recognition

Face Recognition is used to identify people within an image. The usual pipeline for Face Recognition is by first locating the face, then extracting features from that face, and finally comparing those features with some other stored features which will result in either a match or an unknown in some cases [8].

The main problem with Face Recognition is the lack of consistency within either a series of images or a video. Such

consistency includes the pose of the head, facial expression, feature occlusion, illumination, quality of image, and motion blur [9], [8], [10]. These issues will affect feature extraction in different ways which in the end will result in a decrease in accuracy for Face Recognition. Furthermore, with a larger database of these features, the flaws will be highlighted even greater, and thus will result in less optimal results.

To overcome nonconformity in Face Recognition, different methods are created in order to try to extract features that are as much as possible not affected by the different formats of the face. Although other methods exist, a general way that this was handled was by attempting to alter the image slightly such that each face could be overlapped on another and thus highlighting any feature changes [11], [12].

B. Optical Character Recognition

Optical Character Recognition (OCR) is the process that reads text from an image. The OCR will be used after locating the name of an individual within a frame image so that the text from that name could be read.

The general sequence [13] of extracting text from an image is by first preprocessing the image to attempt to reduce noise, and applying some operations to formalise as much as possible the text in the image. Then, each character is segmented on its own so that features are extracted to classify each character individually. Finally, some post-processing can be done which may include Natural Language Processing (NLP) approaches and dictionary-based.

One of the problems that can occur with an OCR system is the different fonts or handwriting that exist. Other issues include the contrast between the text and the background, and also some characters are hard to distinguish from another which may also raise issues [14]. Furthermore, these problems can emerge more with a lower image or camera quality.

For the OCR section in this report, a consistent and generic font is used which helps in eliminating problems which come with specific fonts or handwriting. Furthermore, to handle the rest of the issues, adequate preprocessing must be done so as to distinguish characters from one another and from the background as much as possible.

C. Scene Detection

Scene detection is where a video is split into different segments where each segment is a group of closely related frames. An example of a scene is an interview segment and another is an advertisement section. For the scope of this research, the type of scene is not a priority since when an individual face is found, it is assumed that it is an interview scene.

In general use, scene detection is used for a quick summary of a video and is an important feature in video analysis [15], [16]. Furthermore, this technology allows a user to better understand the difference between scenes and distinguish between them. In some cases [17], [18], key-frames are also extracted from these scenes which are supposed to represent the collective frames within a scene.

There are a number of ways to detect boundaries between different scenes which include the use of histograms, contours, motion, and clustering techniques [5], [15]. The main issue with scene detection is the different unnormalised scene changes such as a cut, opacity transition, or some other unique transitions.

One solution to this problem is to check which algorithm performs best on the specific type of transitions that may be encountered. For example, Wei Qi et al [19] tested different algorithms and measured the algorithm performance with different video types which include news videos.

III. LITERATURE REVIEW

As described by Gao et al. [15] the first step for video analysis is to section related frames together as scenes. To achieve this, numerous methods are available to detect these scene boundaries [15], [20]. To detect these boundaries, both a histogram difference metric (HDM) and a spatial difference metric (SDM) were used to calculate the variance between the neighbouring frame pair. These two metrics were then used inside a fuzzy c-means clustering (FCM) algorithm so that then the scene boundaries can be identified. To combat, abrupt transitions and gradual transitions, the use of binary patterns were used so that the exact location of the boundaries can be located. Once the video was fully segmented into different scenes, then each scene was classified into either ‘anchorperson shots’ or ‘news footage shots’. Instead of using all the frames in each scene, only a single keyframe was used so that each scene could be classified. For the classification, the GTC method, proposed by Zahn [21], was used, which makes use of vertices in the feature spaces to then create a minimum spanning tree (MST). Furthermore, to filter any outliers, a minimum for ‘anchorperson shots’ was set to 2 seconds, and a minimum for ‘news footage shots’ was set to 10 seconds. The results for a total of 14 news videos were relatively well, with 98.5% precision and a recall rate of 94.21%, while also, as described, being faster than most state-of-the-art algorithms due to their new binary pattern-matching strategy. However, it was argued that not all news videos have the similar temporal structure that was used in the study, and thus a more robust news story segmentation might be required to achieve similar or better results. What was interesting about this study was the usage of the key frames as well as the algorithms which achieved very good results.

A. Scene detection

In the study done by Lupatini et al. [5], three different scene detection methodology categories were tested. The algorithm categories were histogram-based, motion-based, and contour based, which also had different algorithms within them, which helped in clarifying which algorithm performed best. Furthermore, the analysis carried out was done on different videos which were categorised as Film, Documentary, News, and Advertising. Each analysis was carried out for each algorithm and video type pair containing the recall and precision for the ‘Cut’, ‘Diss’, and ‘Cover’. The ‘Cover’ shows the coverage of

the scene, the ‘Diss’ shows the dissolving transitions and the ‘Cut’ shows the cut transitions.

Since this report concerns news analysis, the results in the ‘News’ category are more important here. From the paper, the algorithm named ‘H5’ performed better than the others. Although this, the general algorithm category that performed best of all the cases was histogram based with a very high recall and precision for both cut and dissolve transitions. The algorithm named ‘H5’ was based on a global histogram and evaluated on 9-bit colour code information. The best-evaluated algorithm for all video categories was named ‘H7’ which is also based on a global histogram, but instead evaluated on hue information with 256 bins for each histogram.

The histogram method detects a boundary whenever the difference between two frames exceeds a certain threshold. Since motion may trigger this threshold detection, Nagasaka [22] suggested using histograms on the individual colour channels themselves rather than the luminance information only since this may sometimes be deceiving. Since in some cases between boundaries the colours do not shift drastically, Nagasaka [22] also suggested splitting the frame into a 4 by 4 grid when calculating the change. Addressing the issue of the opacity transition, known as dissolve, Zhang [23] suggested using a method called the ‘twin comparison’ technique. This method used two thresholds, a low and a high. Whenever the high threshold is exceeded, a cut transition is detected, and whenever the low threshold is exceeded, a new frame is declared as a candidate for the starting frame of the gradual transition. The starting frame of the gradual transition is then compared with the next frames until the high threshold is exceeded noting the end of the transition and the start of a new scene. However, it is noticed that this technique specifically addresses the opacity transitions and not other unique transitions that may be used within a video.

B. Scene detection – News videos

Qi et al. [19], had a different approach towards scene detection specifically for news videos. In their paper, they used both audio and frame images to segment the news video into shots. Furthermore, NLP was used to classify the video segments into categories for a more contextual description. The structure of the video described in their paper was such that each video contained stories, where each story had a group of shots. Each shot, or scene, was segmented by detecting boundaries where both visual and audio acknowledge a common boundary. The audio was also used as a classification tool where each scene was classified into music, environment sound, silence, or speech, while also further classifying each speaker. Furthermore, results such as 98% were achieved in Lu et al. [7]’s study, which only focused on the audio part of the analysis. On the other hand, visual detection was done using colour correlation analysis which detected shots. Furthermore, using the work of Shahraray et al. [24], key-frames so that not all the frames in the shot would need to be used if the anchor person was detected. Through the papers, [19] and [7] audio

data can also be used for both segmentation and identifying the speaker.

C. Face Recognition

Face recognition is a computer vision technique used to identify faces within an image which can either be a static image or a frame within a video. This identification can be done using different methods, which usually include some sort of feature selection from the image, normalisation so that each of the features could be compared, and finally, a similarity score which describes the likeliness of the person in question with each other person within the normalised database.

As described in [25] face recognition gives rise to multiple issues while considering the noise from the real world. Such issues could arise due to different light levels, facial expressions, head pose, hairstyles or facial hair, and age. To combat these issues, it may be needed to use features that would be unaffected by such issues for example the Quotient Image [26] would be unaffected by light changes. Other solutions would be to normalise the image as much as possible and include more variation so that each edge case could be covered thoroughly.

Recognising faces in videos can be a different method than selecting and reading a single image/frame. An approach used by Lisena et al. [27] utilised an MTCNN to align all the faces and then used FaceNet [28] to extract face embeddings as features. These features were then classified using an SVM which in turn gave a likelihood of a match for each person within the SVM structure. To handle the unknowns, all the face encodings were kept at the end of the analysis so that these features could be inputted into a new model and used for future video analysis. To choose the features that went into the new model, hierarchical clustering was used along with a distance threshold, so that similar encodings and side-facing faces would be filtered from the input to the model.

IV. DESIGN

In this chapter, the design of the system from a high-level view is discussed. First, an overview of the system will be given which should explain how the different components interact with each other. Furthermore, the choices for each component are explained with respect to the literature discussed in Chapter III.

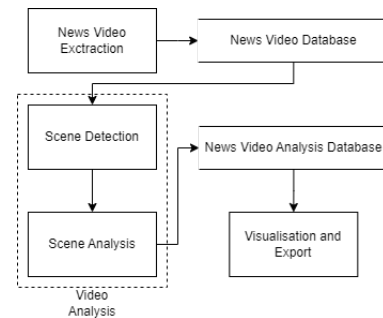


Fig. 1. Component Interactions

A. Overview

Referring to the main goals of this report, discussed in section 1B, 4 main components should be created. Firstly, the *News Video Extraction* component, should be able to extract news videos from the TVM website. Secondly, the *Scene Detection* splits the video into different scenes so that interviews could be easily split from other scenes. Thirdly, the *Scene Analysis* component is a system which retrieves the individual along with the found name if any. This is done with the help of *Face Recognition* which extracts facial features from frames and matches them with a database and also the OCR for *name extraction* which reads the individual names from a given frame or image. This is the main component which creates the timeline which shows the occurrence of an individual per scene. Finally, the *Visualisation and Export* component should show this timeline in an understandable format to the user. Optionally, this final component should also allow the user to export the current database to an external one and also to change individual names if need be. An overview of the interactions of these components can be seen in Figure 1

The generated files from this system would be the analysis files, which include JSON files so that the content would be readable. The analysis files include some information about the video, parameters used for analysis, and most importantly the timeline itself. Furthermore, apart from the analysis files, the database would also need to be stored so that this could be used for Face Recognition.

B. Components

The first component is *News Video Extraction*. This will be done with Web Scraping which is a process where a program extracts data from websites. Once enough videos are collected, the *Scene Detection* component can split the video into different scenes using histogram-based approaches since as seen in III-A these are relatively reliable. For every scene, then the *Scene Analysis* component can utilise both *Face Recognition* and *name extraction* so that individual occurrences can be mapped and thus create the timeline. With the timeline created, the *Visualisation and Export* component will be done with GUI so that the user can easily view the occurrences and export any individuals to an external or new database. The point of this export is so that with each video analysed, the database can increase in size while adding any important individuals for each video analysis.

V. IMPLEMENTATION

In this chapter, the details of the implementation will be described which are based on the design of Chapter IV. First, a general overview of the system will be given so that the flow of components could be better explained. Then the implementation of each of the different components is described including the algorithms and packages used.

A. System Overview

This system should be able to extract news videos from the TVM website, analyse them and finally show a report allowing for the export of the database from the collected data. This analysis should contain the timestamps of individual occurrences along with their names which are also extracted from the video itself.

The whole system is coded in Python language along with various available packages. The reason for choosing this language is the number of powerful packages readily available which can be easily implemented and applied to any program. Another reason for choosing this language is because of the experience level using this language which allows for more efficient and optimised code.

Furthermore, the system was split into 3 main sections which are i) the *News Video Extraction*, ii) *Video Analysis*, and iii) *Visualisation and Export*. Each one of these sections relies on the previous while outputting any files if necessary. The main files created were pickle files and JSON files which contain the exact same data as the pickle files but in a readable format. Moreover, frames of the individuals were also saved as images for better referencing and debugging.

For each video analysis, a *person database* is entered which contains all the individual facial features so that these could be used for the face recognition section. Furthermore, whenever this database was updated within the analysis, this change was only done locally and not updated on the original database. For an external database to be updated, this was to be exported and merged with another existing database, which is automatically created at the end of each video analysis or can be created manually with a separate program.

B. News Video Extraction

The news videos were retrieved as described in the Design (IV) chapter. For this solution, *Selenium* was used due to more experience and reliability with the Python language. Another alternative would have been to use *Puppeteer* but since this works best with *Node.js*, it was decided that it is better to keep everything in Python so that each component could all be linked together with the same language. Both of these packages are web automation tools which allow for automating mouse clicks, reading elements and any other web interactions that would be needed to extract the video from the TVM website.

A total of 53 videos were collected starting from July to September while skipping some days. This was deemed so that proper analysis of the videos could start.

C. Scene Detection - Video Analysis

For Scene Detection, the package *PySceneDetect* was used since this package offers different algorithms which can be used efficiently with segmenting a video into different scenes. The default algorithm was used which calculated the average HSV colour of each frame to calculate the difference between 2 subsequent frames, and if a threshold is exceeded, a boundary would be detected thus starting a new scene.

D. Scene Analysis - Video Analysis

Once the video was split into different scenes, frames were analysed by face recognition and name extract systems so that firstly the individual could be potentially identified, and if not identify a new individual with a name. To continue on this, each frame is analysed separately, and then at the end of the scene averages are calculated to get the correct individual information.

1) *Face Recognition*: First, for each frame face detection was done so that if a single was not found, the analysis could continue without delay. This face detection was done using the Python package *Face Recognition*, which is a package that allows for face detection using the HOG algorithm and also extracts facial features which then can be compared between one another to get a distance metric and check for matches within a database.

During face detection, if a face is detected, then the facial features are extracted along with the name as described in the next section V-D2. Once the facial features are extracted, the same package is used to compare the distances between the *person database* facial features to check whether one matches within a certain threshold. Since matching with features is done for every frame within the scene, an average of the distances is calculated so that the threshold could be calculated on the average distances instead. If there is a match, the scene notes the person match, and if there is no match, the scene notes that a new person was detected and then inputs this person into the local *person database* so that this could be used in the subsequent scenes of the same analysis.

2) *Name Extraction*: The video structure of the data to be analysed all had a similar structure. This structure was such that whenever there was an interview, the name showed up in a white box with blue text.

The name was extracted with the Python package *Pytesseract* OCR. Pytesseract is based on Google's Tesseract-OCR Engine [29] which is one of the best Optical Character Recognition systems currently. For this simple formalised text, there should not be many issues except in small cases such as text transition animations.

E. Visualise and Export

To visualise the analysis results, the Python package *PySimpleGUI* was used, which is a wrapper for *Tkinter*, which is another Python package. These packages allow building GUI programs with a lot of customizability. The reason for choosing *PySimpleGUI* over *Tkinter* is that *PySimpleGUI* is a simpler version which is still as powerful but simplifies a lot of the functionalities. Since this report is not focused on the GUI aspect, *PySimpleGUI* seemed as the perfect tool since it is also based on the Python language.

Through this GUI, the user is able to view the general information about the video such as the duration, name, and size. Furthermore, the timeline of identified people can be shown as well which also allows for the option to export selected individuals to an existing *person database* or create a new one. To continue on this, the GUI allows for the user to

modify the names of the individuals so that the export would be more accurate just in case the OCR failed or in the case of including a more specific name, for example adding the surname. The exported *person database* could then be used as an input for other video analysis as described in V-A

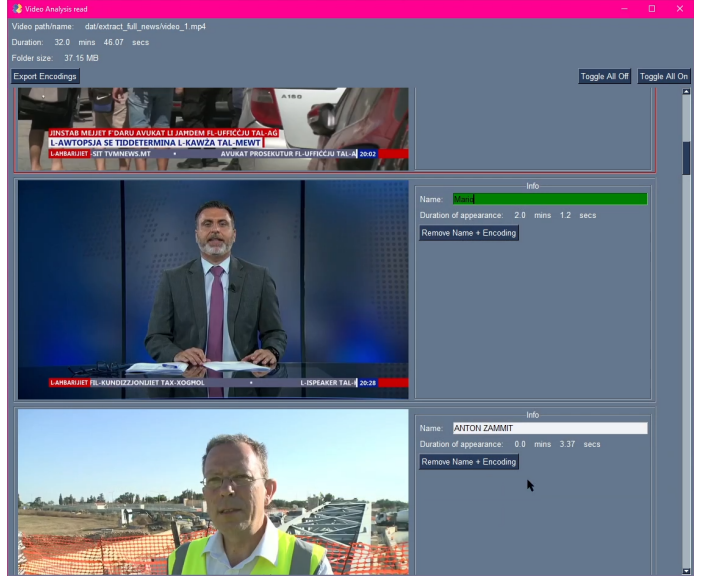


Fig. 2. Current prototype of Visualise and Export

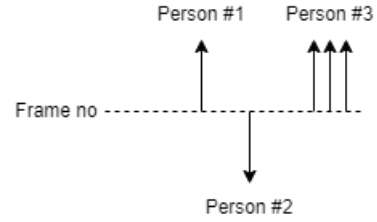


Fig. 3. Person timeline example

F. Reading person timeline

The timeline was read by aligning all the persons found in the timeline one after another. The duration of appearance was calculated by counting the number of coherent frames in which the person appears in. The coherent frames also allowed a tolerance of 5 frames, which allowed a maximum of 5 frames for the person to appear again and still count part of the same duration. Other than the duration, the start and end appearance of the person was calculated by the first and last instances of occurrence for that particular duration.

G. Analysing analysis and creating database

After the analysis for each video, another section was added to export the found faces and encodings into a database to be used in future analysis.

Although this export could be done automatically, it was preferred to have user input to select which faces respective

names to export in the database. The reason for this is that in some cases, some non-interview faces would be identified for example in an advertisement. To add to this, the persons with the highest duration were shown first so that to show which persons are most likely needed in the database.

VI. EVALUATION

The final video analysis can be evaluated by first selecting a sample of the videos and manually labelling each individual occurrence. To compare one another, the individual names can be used to map the differences between the manual labelling and the news video analysis. To continue on this, the accuracy can be calculated as follows:

$$\text{accuracy} = \frac{\text{matches}}{\text{matches} + \text{non-matches}}$$

Note that for this to be clear, the scenes where no faces were detected will not be considered, and only those scenes with facial detection will be considered. Furthermore, the scene timestamps against the manual individual occurrence timestamps will also not be considered at this stage but rather if the individual occurs in that scene, it should be enough.

To evaluate the scene detection along with the face detection an IoU (Intersection over Union) will be used so to highlight any major changes. This will be done using the following:

$$\text{IoU} = \frac{\text{Area}_{\text{Overlap}}}{\text{Area}_{\text{Union}}}$$

Given these metrics, this should give a better insight into how the new video analysis was handled and also how the accuracy changes with a bigger database of facial information.

VII. CONCLUSION

The result of this report aims to demonstrate and create a system which can identify individuals while also extracting their names in an unsupervised manner. From the research, a better understanding of what goes into such a system can be viewed. Furthermore, the architecture of the system can be seen and future improvements and testing will be done to reach an optimal system. This system will be evaluated as described in the evaluation, and also in a time efficient manner, since analysing a video usually takes quite some time, and much more when it comes to multiple videos.

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FYP Gantt chart

